

Bidirectional EV Interconnection Process (VGIC suggested edits, May 2026)

It is the position of the EDCs that electric vehicle ~~charging systems operating in with bidirectional capabilities mode, commonly referred to as V2X, V2G, V2H, and any other terminology referring to export of power from a vehicle to a while grid-connected load~~ and operated in parallel with a power source provided by the utility; will be treated as any other inverter-based battery interconnections as defined in those guidelines within this document, ~~except for the following provisions:~~

- ~~Bidirectional charging systems with an inverter external to the vehicle, referred to as DC Bidirectional Charging Systems, and subject to the guidelines for inverter-based battery interconnection within this document, with the exception that the equipment shall be certified to UL1741 Supplement SB at such time this standard is released, and UL9741 Standard for Safety of Electric Vehicle Power Export Equipment (EVPE).~~
- ~~Bidirectional charging systems with on-board inverter functions in the electric vehicle, referred to as AC Bidirectional Charging Systems, and subject to the guidelines for inverter-based battery interconnection within this document, must either:~~
 - o ~~(a) Be certified as a DER System through the relevant UL 1741 SB Certification Requirement Decision (CRD), or;~~
 - o ~~(b) Use an EV charging station through certified to UL 1741 SC. These requirements are in addition to compliance with Supplements SA and SB otherwise required in these Guidelines for ESS.~~

Special Note:

If the bidirectional charger is integral to the vehicle itself, the vehicle will not be permitted to export to the grid, or offset a load connected in parallel with a grid source of power at any location other than the point of interconnection with Municipal approval, a meter change when required, and the Permission to Operate documentation. Any material modifications to, or replacement of, bi-directional charging equipment will be subject to the requirements as otherwise referenced within these Guidelines.

Commented [ZW1]: VGIC recommends this language instead of "V2X, V2G, V2H," which do not have standard definitions.

Commented [ZW2]: VGIC recommends drawing a distinction here between DC and AC bidirectional charging systems.

DC bidirectional charging systems adhere to IEEE 1547-2018 via UL 1741 SB certification.

AC bidirectional charging systems meet IEEE 1547-2018 through one of two pathways:

1. UL 1741 SB CRD for DER Systems. The vehicle and charger are certified as a single system.
2. UL 1741 SC. The vehicle is designed to SAE J3072, and the UL 1741 SC-certified charger acts as a gatekeeper executing all relevant oversight functions to ensure IEEE 1547-2018 is met.

Commented [ZW3]: i.e., "electric vehicle charging systems operating in bidirectional mode while grid-connected and operated in parallel with a power source provided by the utility"

Commented [ZW4]: UL 1741 SB is applicable to DC Bidirectional Charging Systems.

Commented [ZW5]: Other jurisdictions that have clarified or modified interconnection rules to consider bidirectional charging systems have required UL 1741, but not UL 9741. See, for example, California's Rule 21. Maryland's COMAR 20.50.02. NV Energy's proposed interconnection rules update. NY SIR.

This is because UL 9741 is for one piece of equipment that performs both functions of a UL 1741 Grid Support Inverter and an EV charger. However, some systems may have separate pieces of equipment. E.g., an EVSE wallbox that is a separate piece of equipment from an inverter wallbox.

Commented [ZW6]: i.e., "electric vehicle charging systems operating in bidirectional mode while grid-connected and operated in parallel with a power source provided by the utility"

Commented [ZW7]: Using the UL 1741 SB CRD for DER Systems, UL addresses and certifies the grid interconnection and support functionality directly at the EV. This pathway is best suited for specific installation sites with specific EV and EVSE combinations.

Commented [ZW8]: VGIC recommends removing this, as it is already covered in the above. i.e., DC systems meet UL 1741 SB. AC systems either meet UL 1741 SB via CRD or meet UL 1741 SC, which builds on UL 1741 SB by situating the EVSE as a gatekeeper which only authorizes bidirectional charging from SAE J3072 compatible vehicles.